ORFE 409 Final Examination

Due by 12:00 pm noon on
Friday, December 12, 2025

Please follow the honor code and let this represent your own work.

There are *four* pages here for a total of 100 points.

Chapter 12 of Ross (10 points each): Problems 4, 5, 9, and 13.

Negative Binomial Problem (10 points): Let $\{\mathsf{B}_j(p)\,|j \geq 1\}$ be a sequence of *Bernoulli trials* or an i.i.d. sequence of Bernoulli(p) random variables, where the win probability is

$$p = \mathrm{P}\left\{\mathsf{B}_j(p) = 1\right\} = \mathrm{E}\left[\mathsf{B}_j(p)\right].$$

Now define X to be the *negative binomial* random variable that counts the wins within all the Bernoulli(p) trial winning streaks until the first ℓ losses. This means that we can say that

$$X \equiv \overrightarrow{\mathsf{B}}_1(p) \oplus \cdots \oplus \overrightarrow{\mathsf{B}}_\ell(p),$$

where $\overrightarrow{\mathsf{B}}_j(p)$ is the j-th independent Geometric(p) random variable. Whenever $\ell > 1$, show that:

1. Show that
$$\mathrm{E}\left[1 - \mathsf{B}_1(p)|\, X\right] = \frac{\ell - 1}{\ell - 1 + X},$$

2. Use this result to show that
$$\mathrm{E}\left[\frac{\ell - 1}{\ell - 1 + X}\right] = 1 - p.$$

   Hint: Consider $\mathrm{E}\left[\mathsf{B}_j(p)|\, X = k\right]$ for all integers j ranging from 1 to $k + \ell - 1$.

3. Finally, show that
$$\mathrm{Var}\left[\frac{\ell - 1}{\ell - 1 + X}\right] \leq p \cdot (1 - p).$$

Covariance Problem Part I (5 points): Let A and B be two events. Also let X and Y be two random variables.

1. Show that we have the identity

$$\mathrm{Cov}[A, B] = \mathrm{P}(A \cap B) - \mathrm{P}A \cdot \mathrm{P}B,$$

   by explaining when to treat A and B as events and when to treat them as random variables, i.e. indicator functions.

2. Show that for all events, we have

$$(1 - \mathrm{P}\overline{A} - \mathrm{P}\overline{B})^+ \leq \mathrm{P}(A \cap B) \leq \min(\mathrm{P}A, \mathrm{P}B)$$

3. Show that for all random variables, we have

$$X - Y = \int_{-\infty}^{\infty} \{X > a\} - \{Y > a\} \, da$$

   where the events $\{X > a\}$ and $\{Y > a\}$ are used as indicator functions.

Finance Problem (10 points): Let $\mathfrak{G}$ be a standard Gaussian or a Gaussian(0,1) random variable where its cumulative distribution function is given by Φ.

Write down the *probabilistic* solution for the Black Scholes formula in terms of an expectation of some function of $\mathfrak{G}$, as given in the class notes. We are assuming an initial stock value $S(0)$, a strike price K, an expiration time T, a volatility rate σ^2, and a *non-zero* interest rate r.

Your mission is to use the Girsanov transformation and show that this probabilistic solution equals the standard Black Scholes formula. Hint, use the fact that $x^+ = x \cdot \{x > 0\}$.

Covariance Problem Part II (5 points): If $\widetilde{X}$ and $\widetilde{Y}$ are random variables that simulate the pair X and Y (have the same joint distribution) but are jointly independent of this same pair, then

1. Show that

$$\mathrm{Cov}[X - \widetilde{X}, Y - \widetilde{Y}] = 2 \cdot \mathrm{Cov}[X, Y] = 2 \cdot \mathrm{Cov}[\widetilde{X}, \widetilde{Y}].$$

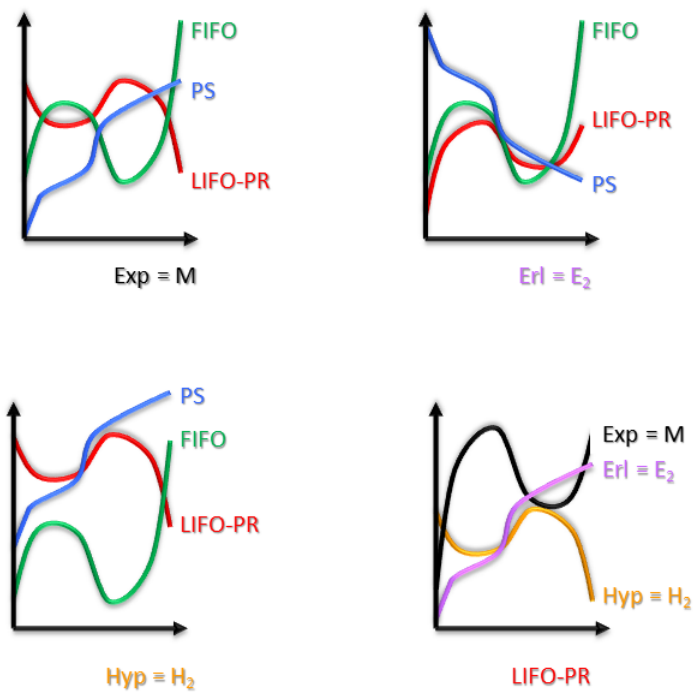
2. Assuming the previous results, show that

$$\mathrm{Cov}[X, Y] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathrm{Cov}[X > a, Y > b] \, da \, db.$$

Queueing Problem (20 points): Simulate the $M/G/1$ queueing system with a time-homogeneous (constant rate) Poisson arrival process using the same three i.i.d. service time random variables as in the midterm (i.e. *Erlang*-2 Gamma(2,2) random variable, *Exponential* (M): an Exponential(1) random variable, and *Hyper-exponential*-2 (H_2): which is the selection of a Exponential(1/9) random variable with probability 0.1 or an Exponential(9) random variable with probability 0.9).

Given an initial queue length $Q(0) = 0$, create the following *four* sets of graphs. They should be formatted like the picture below. Each graph consists of *three* different plots of the mean number $Q(t)$ in the system as a function of time t. Let t range from 0 to 600, where $\mathrm{E}[Q(t)]$ is computed at each time point $k \cdot 0.2$, with $k = 0, 1, \ldots, 3000$:

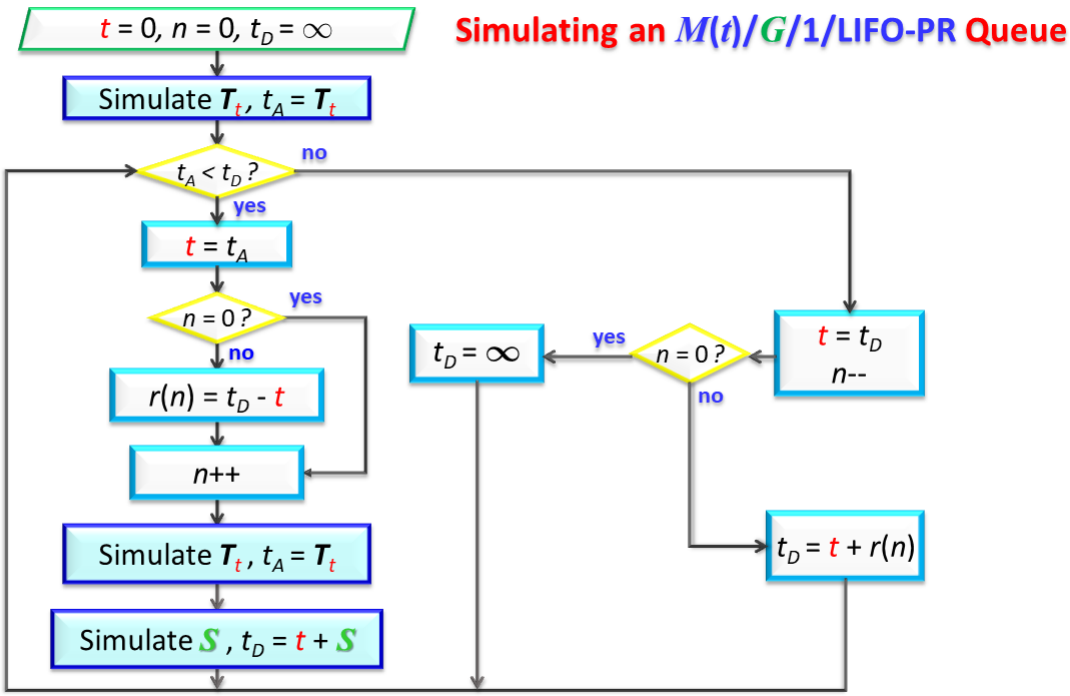
1. For the first group of four graphs, the mean arrival rate is $\lambda_1 = 0.6$. Three of the graphs involve varying the service discipline (one plot for each service discipline) between FIFO, PS, and LIFO-PR (flow chart for this service discipline algorithm is below) for each *one* of the service time distributions E_2, M, and H_2. The fourth graph involves the three service time distributions E_2, M, and H_2 (one plot for each service distribution) but holding the service discipline equal to LIFO-PR.



2. For the second group of four graphs, do the same three plots for each graph but use arrival rate $\lambda_2 = 0.75$.

3. For the third and final group of four graphs, do the same three plots for each graph but use arrival rate $\lambda_3 = 0.9$.

Please title every graph. Also, label the x-axis (time) and y-axis (number of customers), and create a legend box to distinguish each of the three plots.



Covariance Problem Part III (10 points): Assume that F and G are the cumulative distribution functions of X and Y respectively.

1. Show that $F^{-1}(\mathcal{U})$ has the same distribution as X, $G^{-1}(\mathcal{U})$ and $G^{-1}(1-\mathcal{U})$ have the same distribution as Y, and for all real numbers a and b,

   $$\mathrm{Cov}[F^{-1}(\mathcal{U}) > a, G^{-1}(1-\mathcal{U}) > b] \leq \mathrm{Cov}[X > a, Y > b] \leq \mathrm{Cov}[F^{-1}(\mathcal{U}) > a, G^{-1}(\mathcal{U}) > b]$$

   where F^{-1} and G^{-1} are the generalized inverses of the cumulative distribution functions F and G respectively where $\mathcal{U}$ is a standard uniform random variable.

2. Now show that

   $$\mathrm{Cov}[F^{-1}(\mathcal{U}), G^{-1}(1-\mathcal{U})] \leq \mathrm{Cov}[X, Y] \leq \mathrm{Cov}[F^{-1}(\mathcal{U}), G^{-1}(\mathcal{U})],$$

3. Finally, show that the upper bound is always a *positive* correlation, and the lower bound is always a *negative* correlation.

Remark: This shows us how to use simulation to hold the marginal distributions of X and Y fixed but to change the joint distribution. We can then simulate Y to maximize its covariance with X and use its antithetic version to minimize its covariance.